



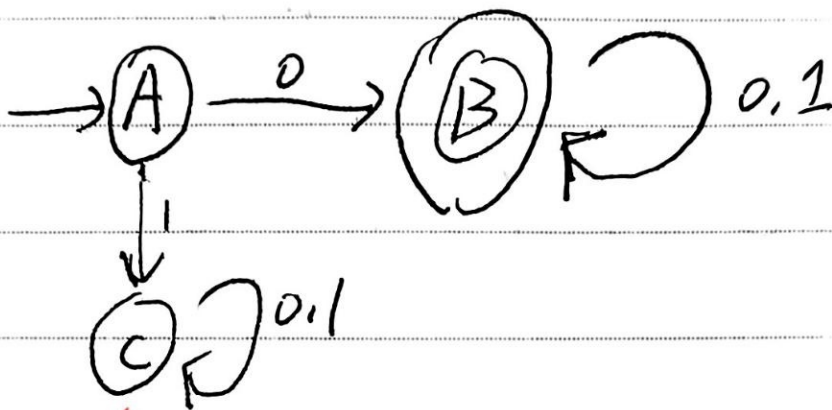
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4. Deterministic Finite Automata (Ex 2)

(Ex 1) $L1 =$ Set of all strings that start with '0'
 $= \{0, 00, 01, 000, 010, 011, 0000, \dots\}$



complete!

Dead State or Trap State

check: eg, 001 $A \xrightarrow{0} B \xrightarrow{0} B \xrightarrow{1} B$ ✓
final state

Eg. 101. initial state $A \xrightarrow{1} C \xrightarrow{0} C \xrightarrow{1} C$ - Not
final state ✗

(Ex 2) Construct a DFA that accepts sets of all strings over $\{0, 1\}$ of length 2.

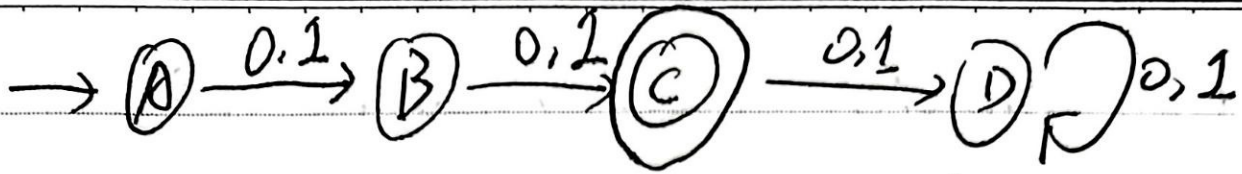
$$\Sigma = \{0, 1\}, \quad L = \{00, 01, 10, 11\}$$



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↑
Dead State or
Trap State

Check: 00

(A) $\xrightarrow{0}$ (B) $\xrightarrow{0}$ (C) - final state. ✓

Eg. 10

(A) $\xrightarrow{1}$ (B) $\xrightarrow{0}$ (C) - final state. ✓

Eg. 001

(A) $\xrightarrow{0}$ (B) $\xrightarrow{0}$ (C) $\xrightarrow{1}$ (D) - X

Eg. 1

(A) $\xrightarrow{1}$ (B) - X



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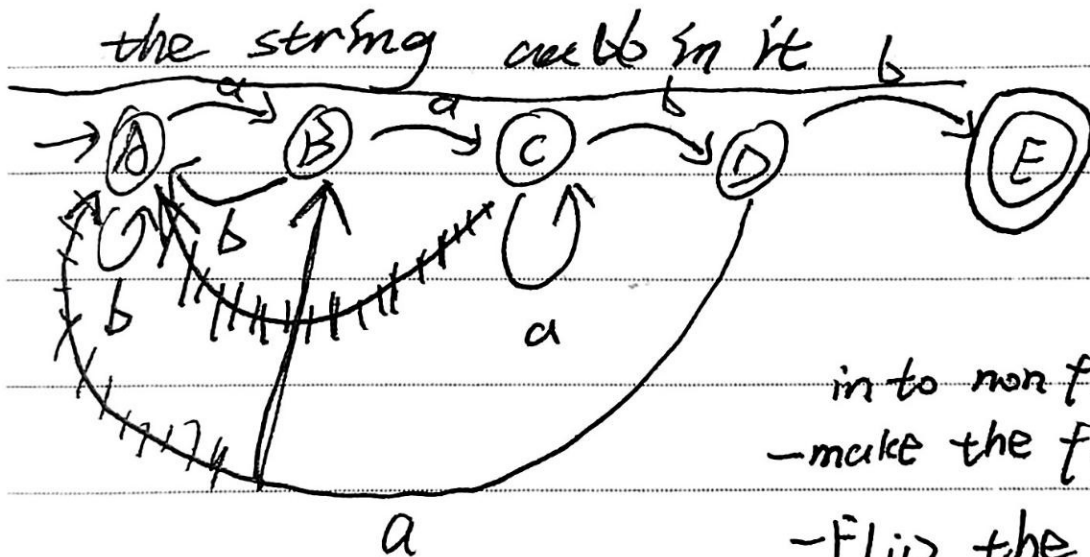
Ex3. Construct a DFA that accepts any strings over $\{a, b\}$ that does not contain the string $aabb$ in it.

$$\Sigma = \{a, b\}$$

Try to design a simpler problem

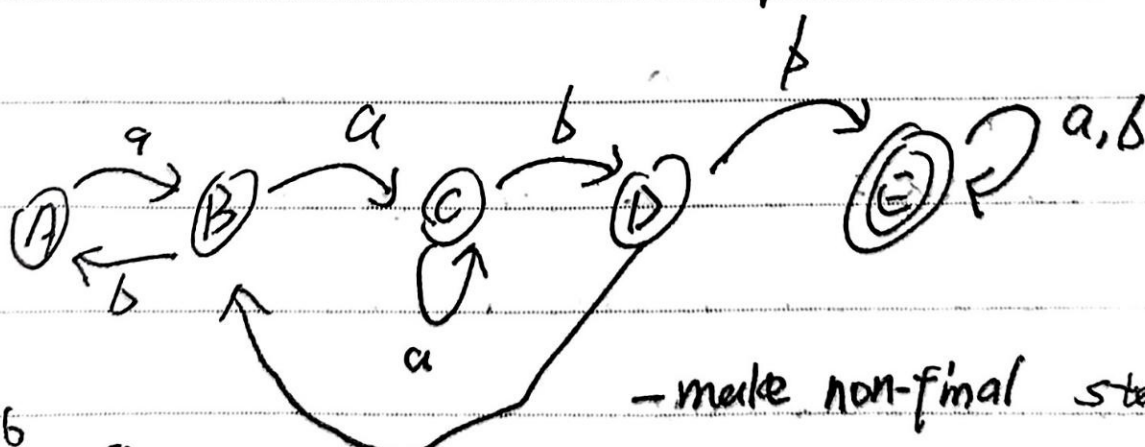
$\Rightarrow$ Let construct - ... that contain

the string $aabb$ in it



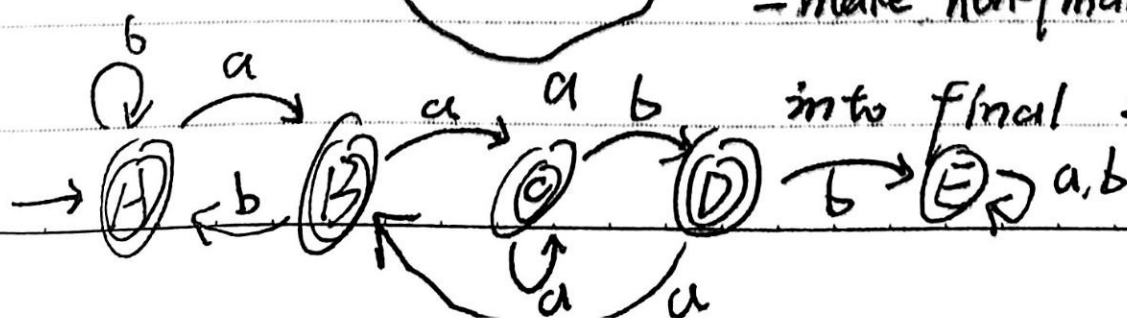
into non final state
- make the final state

- flip the states



- make non-final state

into final state



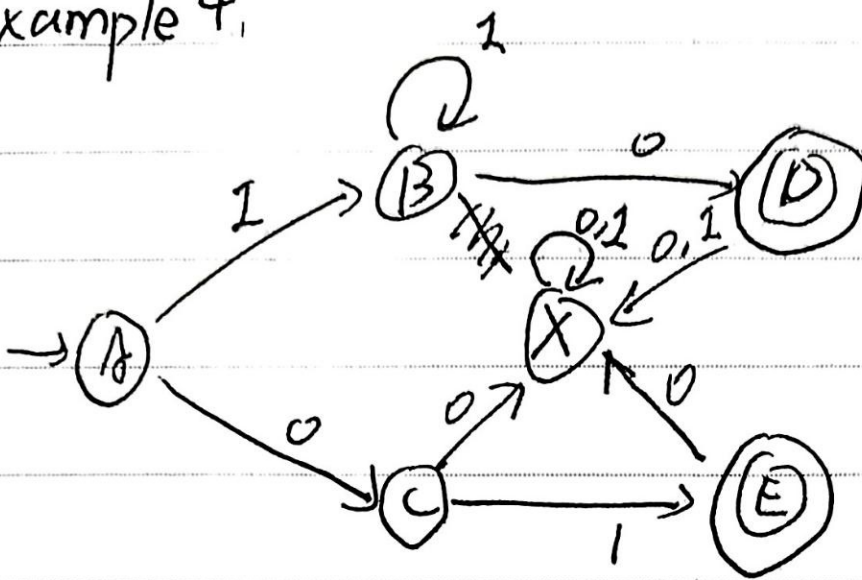


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Example 4.



How to figure out what a DFA recognizes?

10 ✓

01 ✓

21~~0~~ ✓

↑	↑	↑
A	B	D

⇒

$L = \{ \text{Accept the string } 01 \text{ or a string of at least one } 1 \text{ followed by '0'} \}$

Eg. 001, 010, 011, 1101, 1100 X

debt state (X)